

Python Full Stack

₹ 70000/-

Duration: 4 Months

Python

Introduction To Python

- Why Python
- Application areas of python
- Python versions
- Installing python
- Python interpreter architecture
 - Python byte code compiler

Writing and Executing First Python Program

- Using interactive mode
- Using script mode
 - General text editor and command window
 - Idle editor and idle shell
- Understanding print() function
- How to compile python program explicitly

Python Language Fundamentals

- Character set
- Keywords
- Comments
- Variables
- Literals
- Operators
- Reading input from console
- Parsing string to int, float

Python Conditional Statements

- If statement
- If else statement
- If elif statement
- If elif else statement
- Nested if statement

Looping Statements

- While loop
- For loop
- Nested loops
- Pass, break and continue keywords

Standard Data Types

- Int, float, complex, bool, nonetype
- Str, list, tuple, range
- Dict, set

String Handling

- What is string
- String representations
- Unicode string
- String functions, methods
- String indexing and slicing
- String formatting

Python List

- Creating and accessing lists
- Indexing and slicing lists
- List methods
- Nested lists
- List comprehension

Python Tuple

- Creating tuple
- Accessing tuple
- Immutability of tuple

Python Set

- How to create a set
- Iteration over sets
- Python set methods
- Python frozenset

Python Dictionary

- Creating a dictionary
- Dictionary methods
- Accessing values from dictionary
- Updating dictionary
- Iterating dictionary
- Dictionary comprehension

Python Functions

- Defining a function
- Calling a function
- Types of functions
- Function arguments
 - Positional arguments, keyword arguments
 - Default arguments, non-default arguments
 - Arbitrary arguments, keyword arbitrary arguments
- Function return statement
- Map(), filter(), reduce(), any() functions

Modules & Packages

- Why modules
- Script v/s module
- Importing module
- Standard v/s third party modules
- Why packages
- Understanding pip utility

File I/O

- Introduction to file handling
- File modes
- Functions and methods related to file handling

Object Oriented Programming

- Procedural v/s Object Oriented Programming
- OOP Principles
- Defining a Class & Object Creation
- Inheritance
- Encapsulation

- Polymorphism
- Abstraction

Exception Handling

- Difference Between Syntax Errors and Exceptions
- Keywords used in Exception Handling
 - try , except , finally , raise , assert
- Types of Except Blocks
- User-defined Exceptions

GUI Programming

- Introduction to Tkinter Programming
- Tkinter Widgets
- Layout Managers
- Event handling
- Displaying image

Multi-Threading Programming

- Multi-processing v/s Multi-threading
- Need of threads
- Creating child threads
- Functions /methods related to threads
- Thread synchronization

SQL Using MySQL

Introduction to RDBMS

- What is Relational Database Package
- Difference between SQL & Database
- Installing MySQL Server database

SQL Basic

- DDL: Create, Alter, Drop, etc.
- DML: Insert, Update, Delete ,etc.
- DQL : Select
- Autoincrement field
- SQL Comments
- SQL Aliases
- Savepoint & rollback

SQL Constraints

- Not NULL, Unique key
- Primary key, Check
- Default, Foreign key
- Conditional operators
- Like, between, in operators

SQL Clauses

- Order by
- Where
- Limit/top
- Group by
- having

SQL Joins

- Inner Join
- Left Join
- Right Join

- Full Join

SQL View

- creating view
- updating view
- fetching data from view

SQL Functions

- String functions
- Aggregate functions
- Date & time functions

Stored Procedures & Functions

- Understanding stored procedures and their key benefits
- Working with stored procedures
- Studying user-defined functions

Working with CSV Files:

- How to write result to csv file
- How to read csv file

Python Database Connectivity

- Database Drivers and connectors
- Creating connection object
- Understanding cursor object
- Executing SQL statements using cursor
- Fetching records from cursor
- Storing and retrieving Date and Time

Frontend

Introduction to WEB

- What are web applications?
- Static v/s dynamic web applications
- Web application Architecture
 - Front End
 - Back End

HTML

- Introduction to HTML 5
- Parts in HTML Document
- Head Section
- Meta Information
- Body Section
- Heading & Paragraph
- HTML FORMS
- Anchors, Images
- HTML Comments
- HTML Table
- DIV Section

CSS

- Introduction CSS3
- Inline CSS
- Internal CSS

- External CSS
- Styling Text
- Styling Fonts
- CSS Borders
- Selectors
- Backgrounds and Borders
- Text Effects
- Margin & Padding

JavaScript

- What is Script?
- Introduction to JavaScript
- DOM and BOM
- Comments and Types of Comments
- Popup Boxes
- Variables & Operators JavaScript
- Functions and Events Conditional
- Statements Looping Control
- Types of Errors
- Exception Handling
- Java Script Objects
- Browser Objects
- Validations in JS

Bootstrap

- Introduction to Bootstrap4
- Bootstrap CDN & Local
- Bootstrap Classes,Forms,Buttons,Colors
- Bootstrap Grid System

Introduction To React Js

- What is React JS?
- What is SPA?
- DOM vs Virtual DOM
- Advantages and Disadvantages
- Key Features

Environmental Setup

- Node | NPM
- Installation of CLI
- Setup Project
- Directory Structure
- Code Editors
- How React JS Application Boot

Basic Features Of React Js

- React Concepts
- JSX and TSX
- Render Elements
- Function and Class Components
- Props and State
- Handling Events
- Dynamic Data Rendering
- Property Binding

Key Features Of React Js

- Conditional Rendering
- List and Keys
- Forms Handling
- Forms Validations

Component Lifecycle Hook

- Understanding component lifecycle
- All Lifecycle Hooks

Event Handling React

- Understanding React Event System
- Passing arguments to event Handlers

Network Call

- Fetch
- Axios

Custom Services

- Introduction to Services
- Building a Service

Local data storage

- Local Storage
- Session Storage
- Cookies

Routing With React Router

- Setting up React Router
- Configuring route with Route Component
- Making routes dynamic with Route Params
- Working with nested routes
- Link and NavLink
- Redirect Routes

UI Components

- Tailwind CSS

Third Party Libraries

- React Hook Form
- Framer Motion
- React-toastify
- Zod

Introduction To Redux Toolkit(RTK)

- Why Redux
- Install and setup
- Store ,Reducer , actions
- Dispatcher

React Hooks

- Why We Need Hooks.
- Different Types Of Hooks
- Using State And Effect Hooks
- Usereducer , Userref Etc.
- Custom Hooks
- Rules Of Hooks

Deployment

- Deployment on Vercel

Getting Started With Django

- Language v/s Framework
- What is Django Framework?
- Django Versions
- Installing Django

Understanding Django Project

- Creating Django Project
- Creating Django App
- Understanding Directory Structure of Project
- UrlMappings, Views,manage.py
- HttpRequest and HttpResponse Objects
- Registering App in settings.py
- Django Development Server
- Control flow of request processing
- MVT Design Pattern

Django Templates (Presentation Logic)

- Built-in Templates Tags & Filters
- Template Variables
- Template Inheritance
- Building custom template tag
- Integration of Static Contents
 - How to display static images
 - How to use CSS and bootstrap
 - Loading static data to template

Understanding Django Views (Business Logic)

- Why this Component?
- Types of Views
 - Function Based
 - Class Based
- How Views interact with other components
- How to get request data in views
- How to generate response in other formats

Understanding Django Forms

- Why this Component?
- Form class &Field Types
- GET and POST methods
- Form Validations
- How to render django form to template
 - as table
 - as paragraph
 - as li
- How to use Form in views

Django Models (ORM)

- Why this component?
- SQL v/s ORM approach
- Model class and Field Types
- Understanding makemigrations and migrate
- Performing CRUD operations
 - create (insert)
 - read (select)
 - update
 - delete
- Fetching Records from database
 - Understanding QuerySet
 - Fetching all records
 - Fetching records based on conditions
 - Fetching Data in an order
 - Data Slicing
- Understanding Association Mappings
 - one to one
 - one to many
 - many to one
 - many to many
- Integration with MySQL database

Django Admin Interface

- How to create super user
- How to access admin site
- Registering models with admin
- Managing users in the admin

Django State and Session Handling

- Why maintain state?
- Understanding stateless behavior of http protocol
- State management techniques
 - Cookies
 - URL-Rewriting (query strings)
 - Hidden form fields
 - Session
- Understanding Sessions in Details
 - Enable / Disable Session
 - Get and set data with session
 - Using session in template
 - Session Expiry

Django Middleware

- Why middleware?
- Understanding built-in middleware
- Creating custom middleware
 - Function based

- Class based
- How to enable custom middleware

Django User Authentication

- Working with User objects
- Permissions and authorization
- Authentication in web requests
- Managing users in the admin
- Extending the existing User Model

Developing Web Services(APIs)

- What is a Web API?
- Difference between SOAP and RESTful APIs
- Django REST Framework for developing RESTful APIs

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- Difference between SOAP and RESTful APIs
- Django REST Framework for developing RESTful APIs
- Understanding JSON response
- How to consume RESTful API
 - requests package
 - get and post
 - how to parse JSON

Project Deployment (Making Project LIVE)

- Understanding domain and hosting
 - Hosting Providers (AWS)
 - Deploy the project on Amazon EC2
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